Unit 3: Existence, Uniqueness, and Geometry

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Master Unit Reference Guide: Cheat Sheet, Problems, and Solutions

Contents

Unit 3 at a Glance: Core Sub-Topics

This unit is the geometric and theoretical heart of first-order theory. It trades formula-hunting for understanding: how to read an equation as a picture, and why a solution should exist and be the only one.

- **Slope Fields and Vector Fields** — the equation $y' = f(x, y)$ as a field of tiny tangent segments, traced by integral curves.
- **The Phase Plane and Phase Portraits** — states, trajectories, equilibria, and nullclines for a system of equations.
- **Existence and Uniqueness** — the two continuity hypotheses that guarantee exactly one solution through a point.
- **Fixed Points and Iteration** — solving $g(x) = x$ by repeated application, and when the iteration is attracting.
- **The Banach Fixed Point Theorem** — a contraction on a complete space has one fixed point, reached from any start.
- **Picard Iteration** — recasting an IVP as an integral equation and converging to its solution.

Part I — Condensed Cheat Sheet

3.1 Slope Fields and Vector Fields

A first order equation $\frac{dy}{dx} = f(x, y)$ hands you a *number* at every point of the plane, and a derivative is a slope. So the equation is secretly a picture: at each point plant a short segment whose slope is $f(x, y)$. A *solution* is any curve that stays tangent to those segments everywhere it goes.

Slope Field, Integral Curves, and Isoclines

At each point (x, y) the field draws a segment of slope $f(x, y)$.

- An **integral curve** is a solution that is tangent to the field at every point it crosses.
- An **isocline** is the curve $f(x, y) = c$ of all points sharing one slope c; the isocline $f(x, y) = 0$ marks the flat segments.

Worked Example 3.1: The field of $\frac{dy}{dx} = x + y$

Step 1. Read off a slope by substituting coordinates. At $(1, 2)$:

$$f(1, 2) = 1 + 2 = 3.$$

Step 2. Find the flat isocline by setting $f = 0$:

$$x + y = 0 \implies y = -x.$$

Along $y = -x$ every segment is horizontal; integral curves cross it with zero slope.

3.2 The Phase Plane and Phase Portraits

For a *system* $x' = f(x, y)$, $y' = g(x, y)$, the plane no longer carries time on an axis — each point is a complete *state*, and as t runs the state point sweeps out a *trajectory*. A *phase portrait* is a representative bundle of trajectories that reveals the global flow.

Equilibria, Nullclines, and Linear Classification

An **equilibrium** satisfies $x' = 0$ and $y' = 0$ simultaneously. The x-nullcline is $\{x' = 0\}$ and the y-nullcline is $\{y' = 0\}$. For a linear system the eigenvalues of the coefficient matrix classify the origin:

real, same sign	node (source if $+$, sink if $-$)
real, opposite signs	saddle
complex, nonzero real part	spiral
purely imaginary	center (closed orbits)

Worked Example 3.2: Classify $x' = y,\ y' = -x$

Step 1. Set both right sides to zero for the equilibrium:

$$y = 0 \text{ and } -x = 0 \implies (0,0).$$

Step 2. The coefficient matrix $\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ has eigenvalues $\pm i$ — purely imaginary.

$$\text{Purely imaginary} \implies \text{a center: closed orbits around the origin.}$$

3.3 Existence and Uniqueness of Solutions

Before solving, ask whether a solution even exists and whether it is the *only* one. The Big Theorem answers both with two continuity hypotheses — one for each conclusion.

The Existence and Uniqueness Theorem

For $\dfrac{dy}{dx} = f(x,y),\ y(x_0) = y_0$: if f **and** $\dfrac{\partial f}{\partial y}$ are continuous on a rectangle containing (x_0, y_0), then there is a **unique** solution on *some* open interval around x_0. Continuity of f alone delivers *existence*; continuity of $\dfrac{\partial f}{\partial y}$ secures *uniqueness*.

Common Pitfall: Existence Is Not Uniqueness

A continuous f with a *discontinuous* $\frac{\partial f}{\partial y}$ still has a solution — just maybe more than one. For $y' = \sqrt{y}$, $y(0) = 0$, the derivative $\frac{1}{2\sqrt{y}}$ blows up at $y = 0$, and indeed both $y = 0$ and $y = \frac{x^2}{4}$ pass through the origin. The guarantee is also only *local*: never read it as a promise for all x.

Worked Example 3.3: Where uniqueness fails for $\frac{dy}{dx} = \frac{x}{y-2}$

Step 1. Identify where f and $\frac{\partial f}{\partial y}$ lose continuity — watch the denominator:

$$f(x,y) = \frac{x}{y-2} \text{ is undefined when } y - 2 = 0.$$

Step 2. Conclude:

$$\text{The theorem fails along the line } y = 2.$$

Off that line both f and $\frac{\partial f}{\partial y} = -\frac{x}{(y-2)^2}$ are continuous, so a unique solution threads each such point.

3.4 Fixed Points and Iteration

A *fixed point* of g is a value left unmoved by the map: $g(x^*) = x^*$. Geometrically it is where $y = g(x)$ meets the line $y = x$. Feeding outputs back as inputs — $x_{n+1} = g(x_n)$ — either pulls toward such a point or pushes away.

Fixed Point and Stability Test

$$x^* \text{ is a fixed point} \iff g(x^*) = x^*.$$

The iteration $x_{n+1} = g(x_n)$ is **attracting** (stable) at x^* when

$$|g'(x^*)| < 1, \qquad \text{and } \textbf{repelling} \text{ when } |g'(x^*)| > 1.$$

Worked Example 3.4: Fixed points of $g(x) = x^2$

Step 1. Solve $g(x) = x$:

$$x^2 = x \implies x(x-1) = 0 \implies x = 0 \text{ or } x = 1.$$

Step 2. Test stability with $|g'(x)| = |2x|$:

$$|g'(0)| = 0 < 1 \text{ (attracting)}, \qquad |g'(1)| = 2 > 1 \text{ (repelling)}.$$

3.5 The Banach Fixed Point Theorem

The deepest reason iteration converges is *contraction*: a map that shrinks the gap between any two points. On a complete space such a map cannot help but funnel everything to a single point.

Contraction Mapping and Banach's Conclusion

g is a **contraction** with constant $0 \le k < 1$ if

$$|g(x) - g(y)| \le k\,|x - y| \quad \text{for all } x, y.$$

On a **complete** space mapped into itself, Banach guarantees a **unique** fixed point x^*, reached by iteration from *any* start, with the error bound

$$|x_n - x^*| \le \frac{k^n}{1-k}\,|x_1 - x_0|.$$

Common Pitfall: $k = 1$ Is Not Enough

The constant must be *strictly* below one. With $k = 1$ distances need not shrink, so the iterates may never converge — e.g. $g(x) = x + 1$ satisfies $|g(x) - g(y)| = |x - y|$ ($k = 1$) and has *no* fixed point at all.

Worked Example 3.5: $g(x) = \cos x$ is a contraction on $[0, 1]$

Step 1. A bound $|g'(x)| \le k < 1$ makes g a contraction. Here

$$|g'(x)| = |\sin x|, \quad \text{increasing on } [0, 1].$$

Step 2. Its largest value on the interval is at $x = 1$:

$$|\sin x| \le \sin 1 \approx 0.841 < 1.$$

So g is a contraction, and iterating cosine converges to its unique fixed point (the Dottie number, ≈ 0.739) from any start.

3.6 Picard Iteration

Picard iteration is Banach's principle in action on differential equations. First the IVP is recast as an *integral equation*; then repeated substitution builds the solution as a limit.

Integral Equation and the Picard Scheme

The IVP $y' = f(x, y)$, $y(x_0) = y_0$ is equivalent to

$$y(x) = y_0 + \int_{x_0}^{x} f\big(t, y(t)\big)\, dt.$$

Iterate from $\varphi_0 = y_0$:

$$\varphi_{n+1}(x) = y_0 + \int_{x_0}^{x} f\big(t, \varphi_n(t)\big)\, dt.$$

Picard–Lindelöf: if f is continuous and **Lipschitz** in y ($|f(x, y_1) - f(x, y_2)| \le L\,|y_1 - y_2|$), the operator is a contraction, so the solution exists and is unique on some interval.

Worked Example 3.6: Picard iterates for $y' = y,\ y(0) = 1$

Step 1. Start at $\varphi_0 = 1$ and integrate:

$$\varphi_1 = 1 + \int_0^x 1\, dt = 1 + x.$$

Step 2. Feed it back:

$$\varphi_2 = 1 + \int_0^x (1 + t)\, dt = 1 + x + \frac{x^2}{2}.$$

Step 3. The iterates are the partial sums of a familiar series:

$$\varphi_n \to 1 + x + \frac{x^2}{2} + \cdots = e^x,$$

which is exactly the true solution.

Part II — Review Guide: Practice Problems

Work the following ten problems in order. They climb from reading a slope field, through the phase plane and the existence–uniqueness test, into fixed points, contraction mappings, and a full Picard iteration. Solutions follow in Part III.

1. For the slope field of $\dfrac{dy}{dx} = x + y$, find the slope at the point $(1, 2)$ and the isocline along which every segment is horizontal.
2. For $\dfrac{dy}{dx} = y^2 - x$, find the curve along which every field segment has slope 2.
3. For $\dfrac{dy}{dx} = -\dfrac{x}{y}$, find the slope of the segment at $(3, 4)$, and describe the shape of the integral curves.
4. For the system $x' = y$, $y' = -x$, locate the equilibrium and classify it using the eigenvalues of its coefficient matrix.
5. For the linear system $x' = 2x$, $y' = 3y$, write the eigenvalues of the coefficient matrix and classify the origin.
6. For $\dfrac{dy}{dx} = \dfrac{x}{y-2}$, determine where the existence and uniqueness theorem fails to guarantee a unique solution.
7. Show that the initial value problem $\dfrac{dy}{dx} = \sqrt{y}$, $y(0) = 0$ has more than one solution, and explain which hypothesis of the theorem fails.
8. Find every fixed point of $g(x) = x^2$ and classify each as attracting or repelling.
9. Show that $g(x) = \cos x$ is a contraction on $[0, 1]$, and state what the Banach fixed point theorem then concludes.
10. **(Capstone)** For $\dfrac{dy}{dx} = y$ with $y(0) = 1$, carry out Picard iteration from $\varphi_0 = 1$: compute φ_1, φ_2, φ_3, and identify the function the iterates converge to.

Part III — Complete Solutions

Solution 1. The slope is just f evaluated at the point.

$$f(1,2) = 1 + 2 = 3.$$

The horizontal segments lie on the isocline $f = 0$:

$$x + y = 0 \implies y = -x.$$

Solution 2. Set f equal to the target slope and solve.

$$y^2 - x = 2 \implies x = y^2 - 2.$$

Every segment crossing the parabola $x = y^2 - 2$ has slope 2.

Solution 3. Substitute, keeping the leading sign:

$$f(3,4) = -\frac{3}{4}.$$

The integral curves satisfy $y\,dy = -x\,dx$, so $\frac{y^2}{2} = -\frac{x^2}{2} + C_1$, that is

$$x^2 + y^2 = C \qquad \text{(concentric circles).}$$

Solution 4. Equilibria need both derivatives zero:

$$y = 0 \text{ and } -x = 0 \implies (0,0).$$

The coefficient matrix $\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ has characteristic equation $\lambda^2 + 1 = 0$:

$$\lambda = \pm i \text{ (purely imaginary).}$$

Purely imaginary eigenvalues give a center: trajectories are closed orbits circling the origin.

Solution 5. The system is decoupled, so the coefficient matrix $\left(\begin{smallmatrix} 2 & 0 \\ 0 & 3 \end{smallmatrix}\right)$ is diagonal and its eigenvalues sit on the diagonal:

$$\lambda_1 = 2, \quad \lambda_2 = 3.$$

Both are real and positive, so the origin is an unstable node (a source) — every nearby trajectory flows outward.

Solution 6. Locate where f and $\frac{\partial f}{\partial y}$ lose continuity. With $f = \dfrac{x}{y-2}$,

$$\frac{\partial f}{\partial y} = -\frac{x}{(y-2)^2},$$

and both blow up when the denominator vanishes:

$$y - 2 = 0 \implies y = 2.$$

The theorem fails to guarantee uniqueness along the line $y = 2$; everywhere else a unique solution exists.

Solution 7. Here $f = \sqrt{y}$ is continuous at $y = 0$, but

$$\frac{\partial f}{\partial y} = \frac{1}{2\sqrt{y}}$$

is *not* continuous at $y = 0$ — it blows up. So uniqueness is not guaranteed, and indeed two solutions pass through the origin:

$$y = 0 \qquad \text{and} \qquad y = \frac{x^2}{4} \quad (x \geq 0),$$

since $\frac{d}{dx}\left(\frac{x^2}{4}\right) = \frac{x}{2} = \sqrt{x^2/4}$. The failed hypothesis is continuity of $\frac{\partial f}{\partial y}$.

Solution 8. Solve $g(x) = x$:

$$x^2 = x \implies x(x-1) = 0 \implies x = 0,\ 1.$$

Classify with $|g'(x)| = |2x|$:

$$|g'(0)| = 0 < 1 \text{ (attracting)}, \qquad |g'(1)| = 2 > 1 \text{ (repelling)}.$$

Solution 9. A bound $|g'(x)| \leq k < 1$ on an interval makes g a contraction there. With $g(x) = \cos x$,

$$|g'(x)| = |\sin x|,$$

which increases on $[0, 1]$ and so is largest at $x = 1$:

$$|\sin x| \leq \sin 1 \approx 0.841 < 1.$$

Thus g is a contraction with $k = \sin 1$. By Banach, g has a unique fixed point in $[0, 1]$, and iterating cosine from *any* start converges to it.

Solution 10. Use $\varphi_{n+1}(x) = 1 + \int_0^x \varphi_n(t)\,dt$ from $\varphi_0 = 1$.

$$\varphi_1 = 1 + \int_0^x 1\,dt = 1 + x.$$

$$\varphi_2 = 1 + \int_0^x (1+t)\,dt = 1 + x + \frac{x^2}{2}.$$

$$\varphi_3 = 1 + \int_0^x \left(1 + t + \frac{t^2}{2}\right) dt = 1 + x + \frac{x^2}{2} + \frac{x^3}{6}.$$

These are the partial sums of the exponential series, so

$$\varphi_n \to 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \cdots = e^x,$$

the true solution of $y' = y$, $y(0) = 1$.

Unit 3 Key Takeaway

This unit reads differential equations as geometry and guarantees rather than formulas. A slope field turns $y' = f(x, y)$ into a picture its integral curves follow; the phase plane does the same for systems. The existence and uniqueness theorem (f and $\partial f/\partial y$ continuous) says exactly one curve threads each point, and Picard iteration via the Banach contraction principle is the machine that proves it.